

Homework 9

Problem 1 – Hyperspheres (10 points) The surface area and volume of an n -dimensional sphere of radius r may be expressed

$$A = S_n r^{n-1} \quad V = C_n r^n$$

In three dimensions, we know that the area and volume of a “normal” sphere are given by $4\pi r^2$ and $\frac{4}{3}\pi r^3$, respectively, so that $S_3 = 4\pi$ and $C_3 = \frac{4}{3}\pi$. We now seek to develop expressions for arbitrary dimension n .

(a) Express S_n in terms of C_n .

We know that Area is the derivative of Volume. This tells us:

$$\frac{dV}{dx} = nC_n r^{n-1} = S_n r^{n-1} \quad (1)$$

This tells us

$$C_n = \frac{S_n}{n} \quad (2)$$

$$S_n = nC_n \quad (3)$$

$$S_n = \frac{A}{r^{n-1}} \quad (4)$$

$$r^n = \frac{V}{C_n} \quad (5)$$

$$r^{n-1} = \frac{r^n}{r} = \frac{V}{C_n r} \quad (6)$$

$$S_n = \frac{AC_n r}{V} \quad (7)$$

b) To evaluate S_n we use the same trick we used to evaluate the Gaussian integral

$$\int_{-\infty}^{\infty} e^{-\alpha x^2} dx = \sqrt{\frac{\pi}{\alpha}}$$

That is, we multiply together two Gaussian integrals and express the result in polar coordinates:

$$\begin{aligned} \left\{ \int_{-\infty}^{\infty} e^{-\alpha x^2} dx \right\} \left\{ \int_{-\infty}^{\infty} e^{-\alpha y^2} dy \right\} &= \int_{-\infty}^{\infty} dx \int_{-\infty}^{\infty} dy e^{-\alpha(x^2+y^2)} \\ &= \int_0^{\infty} e^{-\alpha r^2} 2\pi r dr = \frac{\pi}{\alpha} \end{aligned}$$

where $S_2 = 2\pi$. Use this same method to show that

$$\int_0^{\infty} e^{-\alpha r^2} S_n r^{n-1} dr = \left(\frac{\pi}{\alpha} \right)^{n/2}$$

We can write out our integral:

$$2\pi \int_0^{\infty} e^{-\alpha(n_1^2 + n_2^2 + \dots + n_k^2)} r^{n-1} dr \quad (8)$$

$$= 2\pi \int_0^{\infty} dn_1 e^{-\alpha n_1^2} \int_0^{\infty} dn_2 e^{-\alpha n_2^2} \dots \int_0^{\infty} dn_k e^{-\alpha n_k^2} r^{n-1} \quad (9)$$

Each of these individually evaluates to $\sqrt{\frac{\pi}{\alpha}}$, so we have the product of n copies of $\sqrt{\frac{\pi}{\alpha}}$.

If I call the variable that is counting the dimension n instead of k , (OOPS)

We have

$$\int_0^{\infty} e^{-\alpha r^2} S_n r^{n-1} dr = \left(\frac{\pi}{\alpha} \right)^{\frac{n}{2}} \quad (10)$$

(c) Solve for C_n and S_n in terms of the gamma function, $\Gamma(n+1) = \int_0^{\infty} x^n e^{-x} dx$. Check your expressions for $n=2$ and $n=3$, and evaluate C_6 and S_6 .

We can represent our function as similar to the gamma function by recognizing that the two forms look similar.

$$\Gamma(n) = \int_0^{\infty} x^{n-1} e^{-x} dx \quad (11)$$

We can write out:

$$S_n \int_0^{\infty} e^{-\alpha r^2} r^{n-1} dr = \int_0^{\infty} x^{n-1} e^{-x} dx \quad (12)$$

We can define

$$ar^2 = x \quad (13)$$

$$r = \left(\frac{x}{a}\right)^{\frac{1}{2}} \quad (14)$$

$$2ardr = dx \quad (15)$$

This lets us substitute into the equation to get

$$s_n \int_0^\infty e^{-x} \left(\frac{x}{a}\right)^{\frac{n-1}{2}} \frac{1}{2ar} dx \quad (16)$$

We can pull out the constants, and this looks like a gamma function:

$$= \frac{s_n}{2a} a^{\frac{1}{2}} \frac{1}{a^{\frac{n-1}{2}}} \int_0^\infty e^{-x} x^{\frac{n-1}{2}} \left(\frac{1}{x}\right)^{\frac{1}{2}} dx \quad (17)$$

$$= \frac{s_n}{2} \frac{1}{a^{\frac{1}{2}} a^{\frac{n-1}{2}}} \int_0^\infty e^{-x} \frac{x^{\frac{n-1}{2}}}{x^{\frac{1}{2}}} dx \quad (18)$$

$$= \frac{s_n}{2} \frac{1}{a^{\frac{n}{2}}} \int_0^\infty e^{-x} x^{\frac{n-2}{2}} dx \quad (19)$$

$$= \frac{s_n}{2} \frac{1}{a^{\frac{n+1}{2}}} \int_0^\infty e^{-x} x^{\frac{n}{2}-1} dx \quad (20)$$

$$= \frac{s_n}{2} \frac{1}{a^{\frac{n}{2}}} \Gamma\left(\frac{n}{2}\right) \quad (21)$$

We therefore have the expression

$$\frac{s_n}{2} \frac{1}{a^{\frac{n}{2}}} \Gamma\left(\frac{n}{2}\right) = \left(\frac{\pi}{a}\right)^{\frac{n}{2}} \quad (22)$$

We can rearrange to get

$$s_n = 2 \left(\frac{\pi}{a}\right)^{\frac{n}{2}} a^{\frac{n}{2}} \frac{1}{\Gamma\left(\frac{n}{2}\right)} \quad (23)$$

$$= \frac{2\pi^{\frac{n}{2}}}{\Gamma\left(\frac{n}{2}\right)} \quad (24)$$

We can rearrange for C_n . We know that

$$c_n = \frac{S_n}{n} \quad (25)$$

$$S_n = nc_n \quad (26)$$

This gives us

$$C_n = \frac{2\pi^{\frac{n}{2}}}{n\Gamma\left(\frac{n}{2}\right)} \quad (27)$$

We can check with $n = 2$ and $n = 3$, since we know $A(2) = 2\pi r$, $V(2) = \pi r^2$ and $A(3) = 4\pi r^2$, $V(3) = \frac{4}{3}\pi r^3$

Our expression yields:

$$A(2) = S_n r^{n-1} \quad (28)$$

$$= \frac{2\pi^{\frac{n}{2}}}{\Gamma\left(\frac{n}{2}\right)} r^{n-1} \quad (29)$$

$$= \frac{2\pi}{\Gamma(1)} r \quad (30)$$

$$= 2\pi r \quad (31)$$

$$V(2) = C_n r^n \quad (32)$$

$$= \frac{2\pi}{2} r^2 \quad (33)$$

$$= \pi r^2 \quad (34)$$

This agrees for 2 dimensions.

$$A(3) = \frac{2\pi^{\frac{3}{2}}}{\Gamma\left(\frac{3}{2}\right)} r^2 \quad (35)$$

$$= \frac{4\pi^{\frac{3}{2}}}{\sqrt{\pi}} r^2 \quad (36)$$

$$= 4\pi r^2 \quad (37)$$

$$V(3) = \frac{4\pi}{3} r^3 \quad (38)$$

These two also work.

Now we can find the parameters for a 6 dimensional sphere.

$$S_n(6) = \frac{2\pi^3}{\Gamma(3)} \quad (39)$$

$$= \frac{2\pi^3}{2} \quad (40)$$

$$= \pi^3 \quad (41)$$

$$C_n = \frac{\pi^3}{6} \quad (42)$$

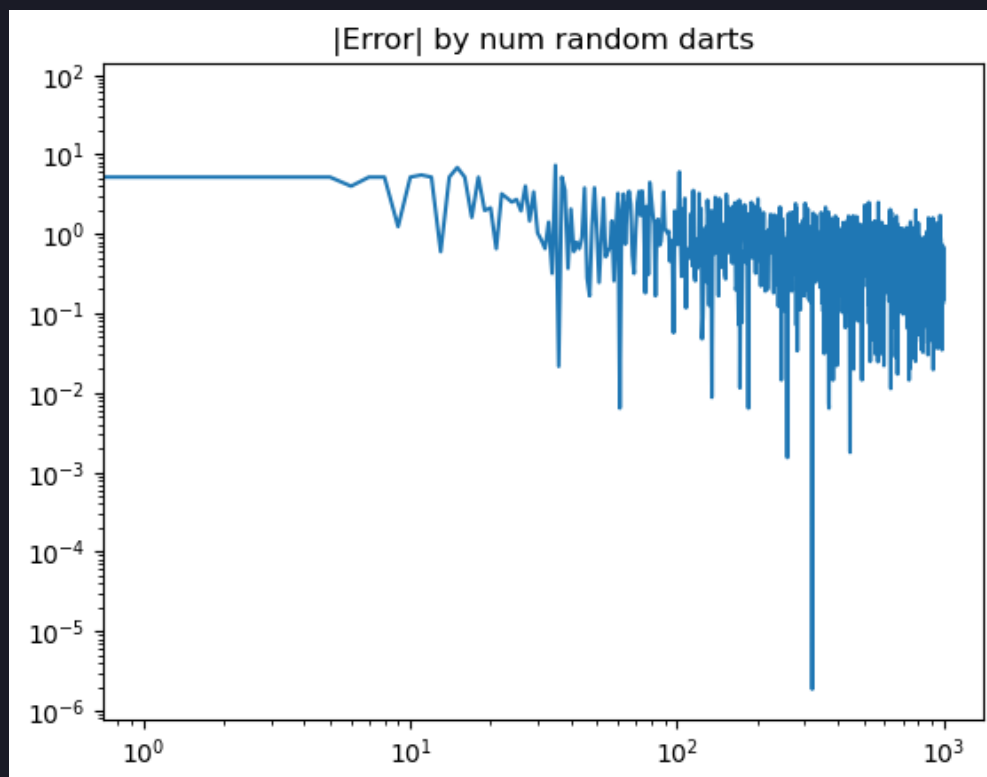
(d) Use a Monte Carlo method to estimate C_6 . Make a log-log plot of the absolute value of the difference between the numerical estimate and the analytic value as a function of the number N of "darts" you throw. Does the trend agree with expectations?

We can simulate a sphere of radius 1.

I get that the volume of the 6 dimensional sphere is ~ 5.1 .

The coefficient $C = 5.1677$.

We see (for only 1 simulation at each point, so it looks somewhat messy), a trend of decreasing error. I would have to do more simulation to find if it is linear in log log.



Problem 2 – Blackbody Radiation An oven with thick, opaque walls is heated to a uniform temperature T . Through a tiny hole in one wall emerges light of a broad range of angular frequencies ω , which is called **cavity radiation** or more commonly **blackbody radiation**. Its spectrum is given by

$$P(\omega) d\omega = \frac{\hbar}{4\pi^2 c^2} \frac{\omega^3}{e^{\hbar\omega/k_B T} - 1} d\omega$$

where $P(\omega)$ is the power per unit area per unit angular frequency. Planck used a computational trick to obtain this result in 1900, but Einstein took it seriously and used the idea that light energy came in discrete packets (which we now call photons) to explain the **photoelectric effect**. This work earned him the Nobel Prize.

- (a) The spectrum above is reported as a function of angular frequency, but experiments typically measure wavelengths, not frequencies. Of course, $c = \lambda \nu = \frac{\lambda \omega}{2\pi}$. Show that the spectral distribution as a function of wavelength is

$$\mathcal{P}(\lambda) = \frac{2\pi \hbar c^2}{\lambda^5} \frac{1}{\exp(\hbar c / \lambda k_B T) - 1}$$

and confirm that it has the appropriate dimensions.

We can do a change of variables:

$$\lambda = \frac{2\pi c}{\omega}, \quad (43)$$

$$\omega = \frac{2\pi c}{\lambda} \quad (44)$$

$$(45)$$

$$d\lambda = -\frac{2\pi c}{\omega^2} d\omega \quad (46)$$

$$\frac{-\omega^2 d\lambda}{2\pi c} = d\omega \quad (47)$$

$$(48)$$

We can define

$$P(\lambda) d\lambda = P(\omega) d\omega \quad (49)$$

$$P(\lambda) d\lambda = \frac{\hbar}{4\pi^2 c^2} \left(\frac{2\pi c}{\lambda}\right)^3 \frac{1}{e^{\frac{(2\pi c \hbar)}{\lambda k_B T}} - 1} \frac{-\left(\frac{2\pi c}{\lambda}\right)^2}{2\pi c} d\lambda \quad (50)$$

$$= -\frac{\hbar (2\pi c)^2}{\lambda^5} \frac{1}{\exp\left(\frac{2\pi c \hbar}{\lambda k_B T}\right) - 1} d\lambda \quad (51)$$

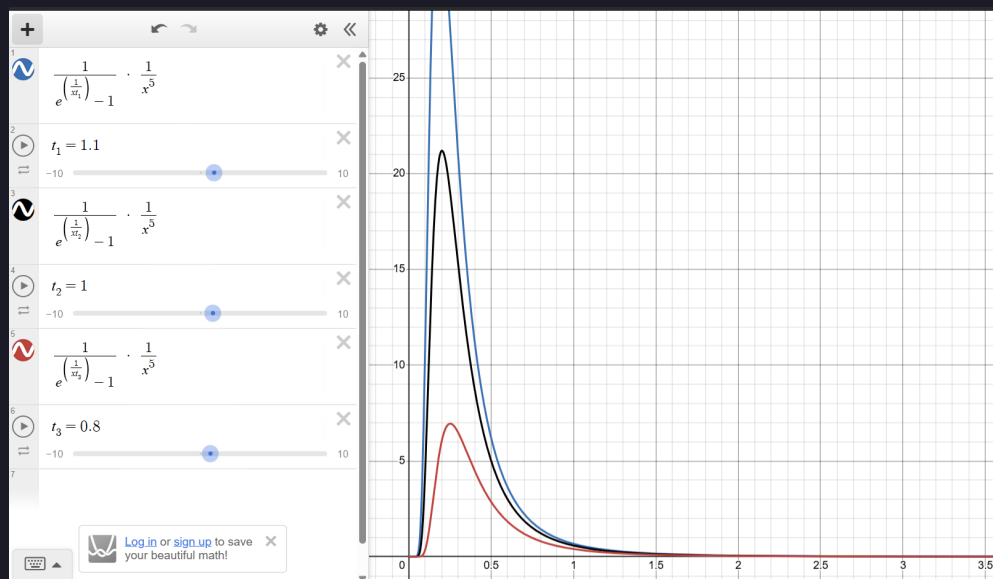
$$P(\lambda) d\lambda = -\frac{2\hbar \pi c^2}{\lambda^5} \frac{1}{\exp\left(\frac{2\pi c \hbar}{\lambda k_B T}\right) - 1} d\lambda \quad (52)$$

(b) Show that if one is only able to measure the long-wavelength portion of the spectrum coming from a distant star, it is impossible to determine the star's surface temperature. *Think very carefully; this is physics, not math!*

Plotting the power (without extra constants) as just

$$\frac{1}{\lambda^5} \frac{1}{e^{\frac{1}{\lambda T}} - 1} \quad (53)$$

gets this power distribution:



(with 3 different "T" values).

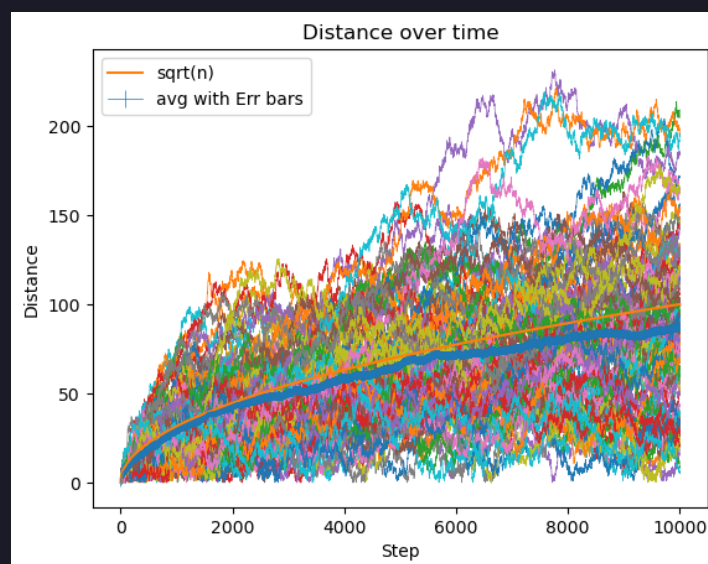
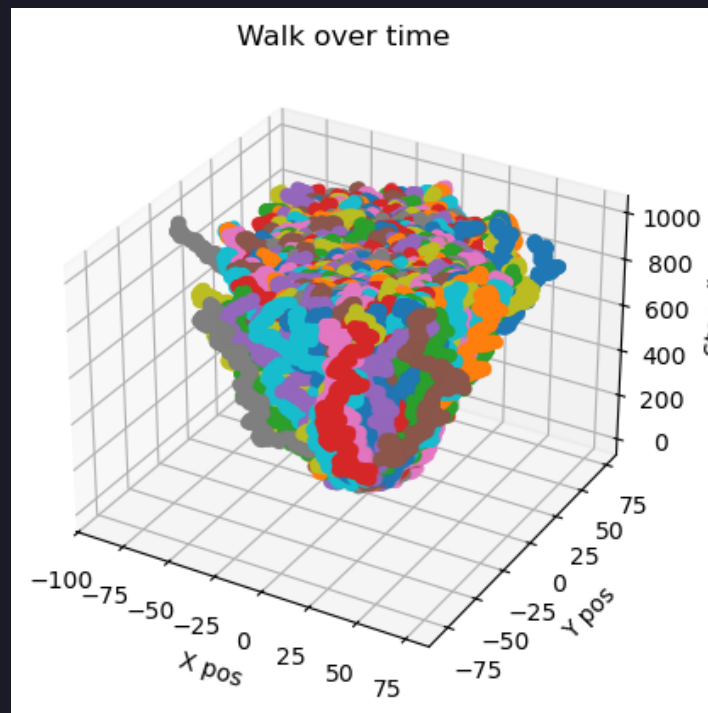
This shows that long wavelengths correspond to relatively low power density - so if we only measure those long wavelengths we are missing most of the information. Even very different temperatures will converge to similar power levels at long wavelengths, so we can't distinguish between them well. Thus, long wave would be pretty useless for getting temperature information (unless the constants dramatically shift the locations of the peak and the spacing between power values at those high wavelengths).

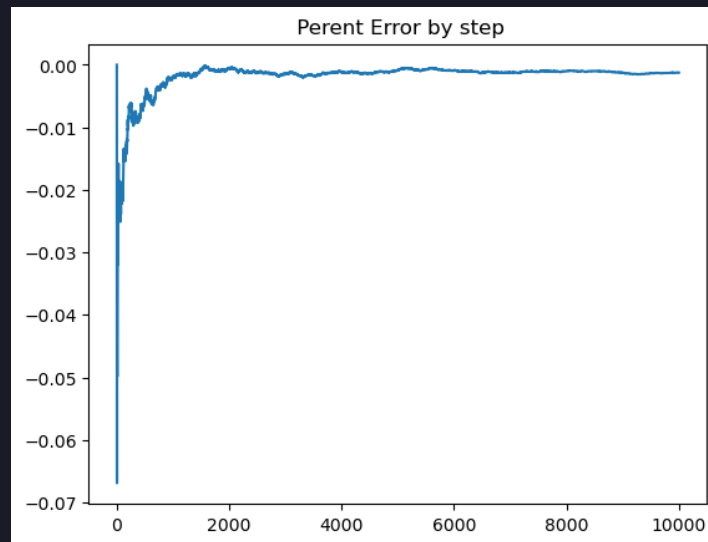
Problem 3 – Two-Dimensional Random Walk A random walker starts at the origin and takes steps of length 1 in a random direction in the xy plane. Simulate the progress of such a walker and compare your results to the expectation that the walker's typical distance from the origin after n steps is $\sqrt{n}$. Use your judgment on how best to make this comparison, but please explain your choices.

I generate a list of 1000 walks that go for 1000 steps each, and average their distances to compare with the predicted curve. I assume that this should be enough to get a good understanding about

the overall behavior, even with all the randomness. I am assuming that the random function I choose acts randomly - `random.sample()`.

The average distance appears close to the prediction, but error bars ($SEM = \frac{\sigma}{\sqrt{n_{\text{random walks}}}}$) are really small and don't encapsulate the curve at the high step limit. Note though - the error relative to the number of steps is basically zero. That means the $\sqrt{n}$ approximation is highly accurate.





Code:

```
import numpy as np

import random as rd

import matplotlib.pyplot as plt
def getDist(x,y):

    return np.sqrt(x**2 + y**2)

getDistv = np.vectorize(getDist)

def randomWalk(numSteps):

    positionsList = np.array([[0,0]])

    nArr = np.arange(0,numSteps+1,1)

    for i in range(numSteps):

        step = rd.sample([[-1,0],[1,0],[0,-1],[0,1]],1)

        newCoordX = positionsList[-1][0] + step[0][0]

        newCoordY = positionsList[-1][1] + step[0][1]

        newCoords = np.array([[newCoordX,newCoordY]])

        positionsList = np.append(positionsList,newCoords,axis=0)
```